

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)			They {have the same <u>genus</u> / are both <u>Apis</u> }(1) They {are closely related/ have similar <u>genes</u> / are similar/ are similar species}(1)		2		2		
	(b)	(ii)		5%/5.0%/4.97%/4.970%/4.9697/ 4.969696969697% = 2 marks Accept 4.9696 recurring If answer incorrect or incorrect rounding accept $82/1650 \times 100$ or $4.96 = 1$ mark		2		2	2	
	(c)	(i)		Any one (x1) from - Using pesticides causes more (winter) losses - Pesticides do not reduce percentage (winter) loss - Pesticides are harmful to bees 2014/2015 losses are the same/ no difference between losses in 2014 and 2015			1	1		1
		(ii)		Any two (x1) from: - Survey only involves {small number/ 5%} of beekeepers/ small sample size/ not enough beekeepers took part - only carried out in North Wales/ does not represent other parts of the country/ only in one area - Has not been reproduced by any other people			2	2		2
	(d)			Any four (x1) from: - Mutation (in a gene) that gave rise to resistance <u>to chemical/ pyrethroid / pesticides</u> - allowed some (Varroa) to survive/non-resistant (varroa) died / survival of the fittest - that breed and pass {advantageous / resistance} gene on to next generation/ that /breed and pass the mutation on - natural selection	2	2		4		
				Question 10 total	2	6	3	11	2	3